SYNAPSE-QUANTUM v3.0 — System Architecture & Mathematical Blueprint

17-Node Wetware Cortical Biocomputer, Two-Photon Gerchberg-Saxton Holography & ONNX Heston Stochastic Volatility Engine

1. Executive Architecture Overview

SYNAPSE-QUANTUM couples a 10,240-neuron 3D cortical organoid simulation (R = 360 um) with a 32x32 FFT Gerchberg-Saxton spatial light modulator (SLM), a 60 FPS WebGL2 3D volumetric Gaussian soma and dendritic tract renderer, and an ONNX Runtime Web Liquid State Machine (LSM) readout driving a Gil-Pelaez Heston stochastic volatility option pricer.

Node	Subsystem	Source Module	Verification Status
Node 1	Genomic Structural Redactor	src/nodes/ecosystem_nodes.ts	Verified (Never writes ID)
Node 2	3D Organoid & Dendritic Connectome	src/core-physics/workers/physics.worker.ts	Verified (10,240 somas, 1,800 tracts)
Node 3	Offline A4 SOP & Forensic ELN	src/nodes/ecosystem_nodes.ts	Verified (pdf-lib + watermark)
Node 4	Stateful Biosecurity Sandbox	src/nodes/biosecurity_sandbox.ts	Verified (Shuffled + parameterized)
Node 5	Seeded Forensic Baseline Jitter	src/nodes/ecosystem_nodes.ts	Verified (SHA-256 -> Mulberry32)
Node 6	Parenchyma Thermal Sentinel	src/core-physics/wgsl/organoid_rk4.wgsl	Verified (38.5 C cutoff)
Node 7	Metabolic-Alpha Bps Ledger	src/data-reservoir/heston_lsm.ts	Verified (Hysteresis + DEMO UPI)
Node 8	Two-Photon Gerchberg-Saxton SLM	src/optics/gerchberg_saxton.ts	Verified (32x32 2D FFT)
Node 9	Neuromorphic YouTube Media Gate	extensions/manifest-v3/content_media_gate.js	Verified (Dynamic quiz)
Node 10	Grants.gov Quant/Wetware Scraper	extensions/manifest-v3/background.js	Verified (XSS-safe textContent)
Node 11	Wavefront Artifact Tracer	src/main.ts	Verified (Zernike coordinate log)
Node 12	Cryptographic SHA-256 ELN Audit	src/nodes/ecosystem_nodes.ts	Verified (WebCrypto digest)
Node 13	Fluorophore Emission Shaders	src/core-physics/workers/physics.worker.ts	Verified (GLSL ES 3.00 4-palette)
Node 14	WGSL Heap Reclaimer	extensions/manifest-v3/background.js	Verified (Protects active/pinned)
Node 15	Autonomous Spike-Reel Daemon	extensions/python-daemon/spike_reel_daemon.py	Verified (FFmpeg 1080x1920 9:16)
Node 16	Adversarial Viva Examiner	src/nodes/biosecurity_sandbox.ts	Verified (6-claim rubric)
Node 17	Cleanroom Console Orchestrator	index.html, src/main.ts	Verified (SAB + Snapshot fallback)

2. Biophysical & Optical Equations

2.1 Hodgkin-Huxley RK4 Membrane & Gating Equations: Each neuron state vector (V_m, m, h, n) is integrated over 20 substeps of h = 0.05 ms per frame using 4th-order Runge-Kutta (RK4) in WebGPU (src/core-physics/wgsl/organoid_rk4.wgsl) with Q_10 = 3 temperature scaling phi = 3^((T - 37)/10). Intracellular ATP modulates Na+/K+ reversal potentials E_Na = 50 * pump(ATP) mV and E_K = -77 * pump(ATP) mV.

2.2 Two-Photon Ballistic Scattering & 4-State ChR2 Photocycle: Two-photon excitation scales quadratically with laser power P_laser and attenuates with depth z below the organoid surface (z_top = -360 um) as exp(-2 * z / l_s) with scattering mean free path l_s = 180 um, multiplied by axial Gaussian confinement around slm_focal_z and the 32x32 Gerchberg-Saxton hologram intensity mask.

2.3 Heston Gil-Pelaez Option Pricer & Feller Condition: The 64-bin spatial reservoir (4x4x4 grid with log-spaced leaky integrator time constants tau_k in [0.2 s, 5.0 s]) feeds a standardized ridge readout exported to ONNX Opset 13 (public/models/heston_readout.onnx). Vol-of-vol sigma is clamped to 0.99 * sqrt(2 * kappa * theta) so the Feller boundary condition 2 * kappa * theta > sigma^2 holds strictly.